

COMSATS University Islamabad

Synopsis for MS Ph.D.

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Proposed Title of the Thesis:

Association of VDR Gene Variability, Fluoride Exposure and Fluorosis Among Children
Population of Thal Desert Areas.

Supervisory Committee

Name and Designation	Role
Dr. Ismat Nawaz (Associate Professor)	Supervisor
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	Member
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Summary of the Research

In many parts of Pakistan, especially among children, the problem of fluoride contamination in groundwater is an unresolved issue of concern to the entire population. The constant consumption of fluoride in excess of the standards will result in skeletal fluorosis, a disease that causes the discolouring of enamel, malformed bones, and decreased mineralisation. Although environmental exposure levels are similar across individuals, the degree of fluorosis

manifestation varies, suggesting that genetic diversity can affect fluoride metabolism and toxicity. The Vitamin D Receptor (VDR) gene is one of the genetic determinants currently being studied; this gene is important in calcium absorption and bone homeostasis. The biological response to fluoride exposure might be dependent upon polymorphisms in this gene, especially the single-nucleotide polymorphism (SNP) rs731236 (TaqI), which alters receptor activity and downstream signalling. The current research is planned as a cross-sectional molecular study of school-going children living in fluoride-exposure regions of Punjab, including Thal Desert. Analyses of water and biological samples will be conducted to determine the degree of fluoride exposure, while genotyping of the VDR TaqI SNP will be performed using PCR-based molecular techniques. Statistical testing of genotype distribution and fluorosis severity will be performed to establish potential gene–environment interactions. The results should increase the body of knowledge about fluoride susceptibility at the molecular level, establish genetic predispositions, and help design specific prevention interventions in susceptible groups.

1. Introduction

Fluoride is a naturally occurring element in the earth's crust and enters groundwater systems at highly variable concentrations, reflecting regional geology and hydrochemical conditions. Within optimum doses, fluoride improves dental health by enhancing enamel mineralisation and preventing dental cavities (WHO, 2017). Excessive consumption, however, causes severe health effects collectively known as fluorosis. Dental fluorosis results from excess fluoride during tooth development, causing enamel defects and discolouration; the skeletal form accumulates fluoride in bone tissue, causing pain, stiffness, and skeletal deformities.

Pakistan records many areas with groundwater fluoride above the WHO reference of 1.5 mg/L. Machine-learning models applied to more than 5 000 fluoride readings estimated that approximately 13 million Pakistani residents live in regions with groundwater fluoride exceeding 1.5 mg/L (Khan et al., 2023). Children are particularly sensitive because developing bone and dental structures incorporate fluoride more readily than adult tissue.

Despite similar environmental exposures, individual differences in fluorosis severity are highly variable, indicating that genetic predisposition may be a major modifying factor (Wang et al., 2022). The VDR gene is a strong candidate: its protein is a nuclear receptor for 1,25-dihydroxyvitamin D that controls calcium absorption, bone mineralisation, and skeletal homeostasis (Holick, 2006). VDR polymorphisms can alter receptor expression and activity, thereby changing calcium metabolism and the individual response to fluoride exposure. The TaqI SNP (rs731236), situated at the 3' end of the VDR gene, is among the most investigated. Its genotypes have been associated with changes in bone mineral density and mineral metabolism across diverse populations (Górczyńska-Kosiorz et al., 2024). Despite these advances, the literature examining VDR TaqI (rs731236) and skeletal fluorosis in Pakistani children remains sparse, a gap this study aims to fill.

2. Literature Review

Endemic fluorosis is caused by chronic intake of high-fluoride water, brick-tea, or salt, resulting in dose-dependent skeletal and dental phenotypes. Host genetics confer additional control over who actually develops fluorosis, particularly among children whose enamel and bone are actively mineralising (Pramanik & Saha, 2017; Niazi & Al-Khanati, 2023).

Early gene–environment research in high-fluoride areas demonstrated that risk is altered by detoxification and bone-matrix genes such as *GSTP1* and osteocalcin, inspiring candidate-gene analyses in fluoride-exposed populations (Wu et al., 2015). In brick-tea regions of China, *GSTP1* polymorphisms were among the first to be linked to skeletal fluorosis, implicating oxidative-stress pathways. Subsequent paediatric work in high-fluoride areas investigated VDR FokI patterns in children, relating VDR genotypes to fluoride exposure conditions—an early indication that VDR biology could determine fluoride sensitivity in development (Yang et al., 2011).

A larger cross-sectional case-control study later identified VDR FokI CT/TT genotypes as associated with reduced risk of skeletal fluorosis in brick-tea regions (Yang et al., 2016). Enamel-matrix genes were also found to be relevant: *COL1A2* rs412777 was associated

with dental fluorosis severity, placing matrix/mineralisation genetics as central to fluoride responses (Jarquín-Yñezá et al., 2018). A systematic review of SNPs and dental fluorosis confirmed multiple implicated loci (matrix, signalling, antioxidant), supporting the view that genetic background contributes substantially to inter-individual variability (González-Casamada et al., 2022).

The TaqI rs731236 variant has demonstrated case-control differences in caries-focused cohorts (Antunes et al., 2016; Lei et al., 2021), indicating that VDR signalling plays a functional role in enamel mineral homeostasis—the same axis activated during fluoride responses in amelogenesis. A meta-analysis of VDR polymorphisms (FokI, TaqI, ApaI, BsmI) confirmed they mediate oral risk in children (Peponis et al., 2023), suggesting that rs731236-mediated modulation of fluoride sensitivity is biologically plausible, although direct paediatric fluorosis data remain limited.

More recent studies continue to implicate alternative genes: COL1A2 variants were linked to dental fluorosis in Tunisian and Latin-American cohorts (Kallala et al., 2024; Jarquín-Yñezá et al., 2018), while AQP5, MMP2, and COMT polymorphisms were investigated in Brazilian endemic fluorosis populations (de Lavôr et al., 2025). These findings support a polygenic model of fluoride susceptibility. Thai school children also revealed salivary proteome changes in severe fluorosis (Gavila et al., 2024), and a 2024 study detected genetic instability signals in fluorosis patients, confirming systemic cellular stress from chronic fluoride exposure (García et al., 2024).

A BMC Pediatrics meta-analysis (2024) confirmed the association of rs731236 with caries risk in children and suggested that the 3'-end position of this SNP may interfere with mRNA processing or VDR structure, affecting functional signalling in mineralising tissues (Qin et al., 2024). AI-assisted models now incorporate both genotype and environmental data to predict enamel outcomes (Wei et al., 2025), a framework that could be adapted to fluorosis risk prediction with inclusion of VDR.

In summary, while VDR FokI remains the most directly studied locus in fluorosis, the bio-

logical plausibility of rs731236 (TaqI) involvement is well-supported by parallel caries and bone-density literature. Child-specific, properly powered studies combining TaqMan/NGS genotyping, urinary fluoride measurement, 25(OH)D assessment, and standardised fluorosis indices (TFI/Dean's) are needed to determine whether rs731236 stratifies fluorosis risk/severity in endemic settings.

3. Problem Statement

High fluoride levels in drinking water remain a significant public health concern in Pakistan, where groundwater fluoride frequently exceeds the WHO guideline of 1.5 mg/L. In heavily contaminated regions such as Punjab, dental and skeletal fluorosis have become endemic, manifesting as enamel mottling, defective bone mineralisation, chronic pain, and skeletal deformities. Although fluoride exposure is the primary environmental driver, considerable inter-individual variation in fluorosis incidence and severity exists among people exposed to comparable fluoride levels, implying a genetic component in the response to fluoride toxicity.

Emerging evidence suggests that polymorphisms in the VDR gene—a key regulator of calcium absorption, bone remodelling, and mineral homeostasis—may modulate fluorosis susceptibility. Among these, the VDR TaqI (rs731236) SNP has been hypothesised to influence receptor activity and downstream calcium metabolism, thereby affecting the skeletal response to chronic fluoride exposure. Despite global interest in fluoride–gene interactions, the role of VDR rs731236 in skeletal fluorosis remains understudied, particularly in Pakistan.

This study therefore aims to examine the relationship between VDR TaqI (rs731236) polymorphism and fluorosis severity in children residing in high-fluoride regions of Punjab, combining environmental exposure data with genetic profiling to identify genetically vulnerable subgroups. Findings will inform targeted public health measures and advance understanding of gene–environment interactions underlying fluoride toxicity in vulnerable Pakistani populations.

4. Objectives

- To evaluate fluoride levels in water and to determine the incidence of skeletal fluorosis in children in the region of Punjab, Pakistan.
- To analyse samples at the genetic level for potential single-nucleotide polymorphisms (SNPs) using tetra-primer ARMS-PCR.
- To identify the relationship among fluoride exposure, skeletal fluorosis, and single-nucleotide polymorphisms (SNPs).

5. Research Methodology

5.1 Ethical Consideration

Before initiating the research, formal approval will be sought from the Ethical Review Board (ERB) of the Department of Biosciences, COMSATS University Islamabad. The study will be conducted in the Thal Desert area of Punjab, which has documented high groundwater fluoride levels.

5.2 Population and Sampling Strategy

One hundred children aged 12–16 years (Grades 7–10) will be recruited from schools in the study area using stratified random sampling. Each participant will provide a 5 mL venous blood sample: 2 mL in EDTA vacutainers for DNA extraction and 3 mL in gel vacutainers for serum separation. Serum samples will be centrifuged and stored at -20°C pending biochemical analysis.

5.3 Fluoride Quantification

Fluoride concentrations in drinking water and serum samples will be measured by the ion-selective electrode (ISE) method to obtain true individual exposure levels.

5.4 Genomic DNA Isolation

Genomic DNA will be extracted using the Phenol–Chloroform–Isoamyl Alcohol (PCI) method. Extracted DNA will be assessed by 2% agarose gel electrophoresis and stored at -20°C until further molecular analysis. Specific primers for the VDR TaqI (rs731236) polymorphism will be designed using Primer3 software and validated for specificity, efficiency, and amplification consistency.

5.5 PCR and Gel Analysis / Genotyping

VDR TaqI polymorphism will be genotyped by Tetra-Primer Amplification Refractory Mutation System PCR (Tetra-ARMS PCR). Amplified products will be resolved by agarose gel electrophoresis; homozygous and heterozygous genotypes will be distinguished by fragment size.

5.6 Statistical Analysis

Data will be entered into SPSS (version 22) and GraphPad Prism. Hardy–Weinberg equilibrium will be assessed for VDR rs731236 genotype frequencies. Comparative genotypic and allelic frequencies between fluorosis and control groups will be used to measure associations with disease susceptibility, with statistical significance set at $p = 0.05$. Logistic regression will be applied to identify independent relationships among VDR polymorphism, fluoride exposure, and fluorosis severity.

Tentative Schedule

Activity	Sept-Oct	November	December	January	February	March	April	May-June
Literature review & Ethical Approval								
Sampling								
DNA Extraction & Genotyping								
Fluoride Quantification								
Statistical Analysis								
Thesis Writing								
Submission								

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